

from 21 meteorological stations located in Germany, with a sampling rate of 10 minutes. The Traffic dataset includes data on the occupancy rates of the freeway system, recorded from 862 sensors across the State of California, with a sampling rate of 1 hour. The influenza-like illness (ILI) dataset contains records of patients experiencing severe influenza with complications.

The M4 benchmark comprises 100K time series, amassed from various domains commonly present in business, financial, and economic forecasting. These time series have been partitioned into six distinctive datasets, each with varying sampling frequencies that range from yearly to hourly. The M3-Quarterly dataset comprises 756 quarterly sampled time series in the M3 benchmark. These series are categorized into five different domains: demographic, micro, macro, industry, and finance.

B.3 EVALUATION METRICS

For evaluation metrics, we utilize the mean square error (MSE) and mean absolute error (MAE) for long-term forecasting. In terms of the short-term forecasting on M4 benchmark, we adopt the symmetric mean absolute percentage error (SMAPE), mean absolute scaled error (MASE), and overall weighted average (OWA) as in N-BEATS (Oreshkin et al., 2020). Note that OWA is a specific metric utilized in the M4 competition. The calculations of these metrics are as follows:

$$\begin{aligned} \text{MSE} &= \frac{1}{H} \sum_{h=1}^T (\mathbf{Y}_h - \hat{\mathbf{Y}}_h)^2, & \text{MAE} &= \frac{1}{H} \sum_{h=1}^H |\mathbf{Y}_h - \hat{\mathbf{Y}}_h|, \\ \text{SMAPE} &= \frac{200}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{|\mathbf{Y}_h| + |\hat{\mathbf{Y}}_h|}, & \text{MAPE} &= \frac{100}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{|\mathbf{Y}_h|}, \\ \text{MASE} &= \frac{1}{H} \sum_{h=1}^H \frac{|\mathbf{Y}_h - \hat{\mathbf{Y}}_h|}{\frac{1}{H-s} \sum_{j=s+1}^H |\mathbf{Y}_j - \mathbf{Y}_{j-s}|}, & \text{OWA} &= \frac{1}{2} \left[\frac{\text{SMAPE}}{\text{SMAPE}_{\text{Naive2}}} + \frac{\text{MASE}}{\text{MASE}_{\text{Naive2}}} \right], \end{aligned}$$

where s is the periodicity of the time series data. H denotes the number of data points (i.e., prediction horizon in our cases). $\mathbf{Y}_h$ and $\hat{\mathbf{Y}}_h$ are the h -th ground truth and prediction where $h \in \{1, \dots, H\}$.

B.4 MODEL CONFIGURATIONS

The configurations of our models, relative to varied tasks and datasets, are consolidated in Tab. 9. By default, the Adam optimizer (Kingma & Ba, 2015) is employed throughout all experiments. Specifically, the quantity of text prototypes V' is held constant at 100 and 1000 for short-term and long-term forecasting tasks, respectively. We utilize the Llama-7B model at full capacity, maintaining the backbone model layers at 32 across all tasks as a standard. The term input length T signifies the number of time steps present in the original input time series data. Patch dimensions d_m represent the hidden dimensions of the embedded time series patches prior to reprogramming. Lastly, heads K correlate to the multi-head cross-attention utilized for patch reprogramming. In the four rightmost columns of Tab. 9, we detail the configurations related to model training.

Table 9: An overview of the experimental configurations for TIME-LLM. ‘‘LTF’’ and ‘‘STF’’ denote long-term and short-term forecasting, respectively.

Task-Dataset / Configuration	Model Hyperparameter					Training Process			
	Text Prototype V'	Backbone Layers	Input Length T	Patch Dim. d_m	Heads K	LR*	Loss	Batch Size	Epochs
LTF - ETTh1	1000	32	512	16	8	10^{-3}	MSE	16	50
LTF - ETTh2	1000	32	512	16	8	10^{-3}	MSE	16	50
LTF - ETTh1	1000	32	512	16	8	10^{-3}	MSE	16	100
LTF - ETTh2	1000	32	512	16	8	10^{-3}	MSE	16	100
LTF - Weather	1000	32	512	16	8	10^{-2}	MSE	8	100
LTF - Electricity	1000	32	512	16	8	10^{-2}	MSE	8	100
LTF - Traffic	1000	32	512	16	8	10^{-2}	MSE	8	100
LTF - ILI	100	32	96	16	8	10^{-2}	MSE	16	50
STF - M3-Quarterly	100	32	$2 \times H^\dagger$	32	8	10^{-4}	SMAPE	32	50
STF - M4	100	32	$2 \times H^\dagger$	32	8	10^{-4}	SMAPE	32	50

$^\dagger H$ represents the forecasting horizon of the M4 and M3 datasets.

* LR means the initial learning rate.

C HYPERPARAMETER SENSITIVITY

We conduct a hyperparameter sensitivity analysis focusing on the four important hyperparameters within TIME-LLM: namely, the number of backbone model layers, the number of text prototypes V' , the time series input length T , and the number of patch reprogramming cross-attention heads K . The correlated results can be found in Fig. 6. From our analysis, we derive the following observations: (1) There is a positive correlation between the number of Transformer layers in the backbone LLM and the performance of TIME-LLM, affirming that the scaling law is preserved post-LLM reprogramming.; (2) Generally, acquiring more text prototypes enhances performance. We hypothesize that a limited number of prototypes V' might induce noise when aggregating language cues, consequently obstructing the efficient learning of highly representative prototypes essential for characterizing the input time series patches; (3) The input time length T exhibits a direct relation with forecasting accuracy, particularly evident when predicting extended horizons. This observation is logical and is in congruence with conventional time series models; (4) Increasing the number of attention heads during the reprogramming of input patches proves to be advantageous.

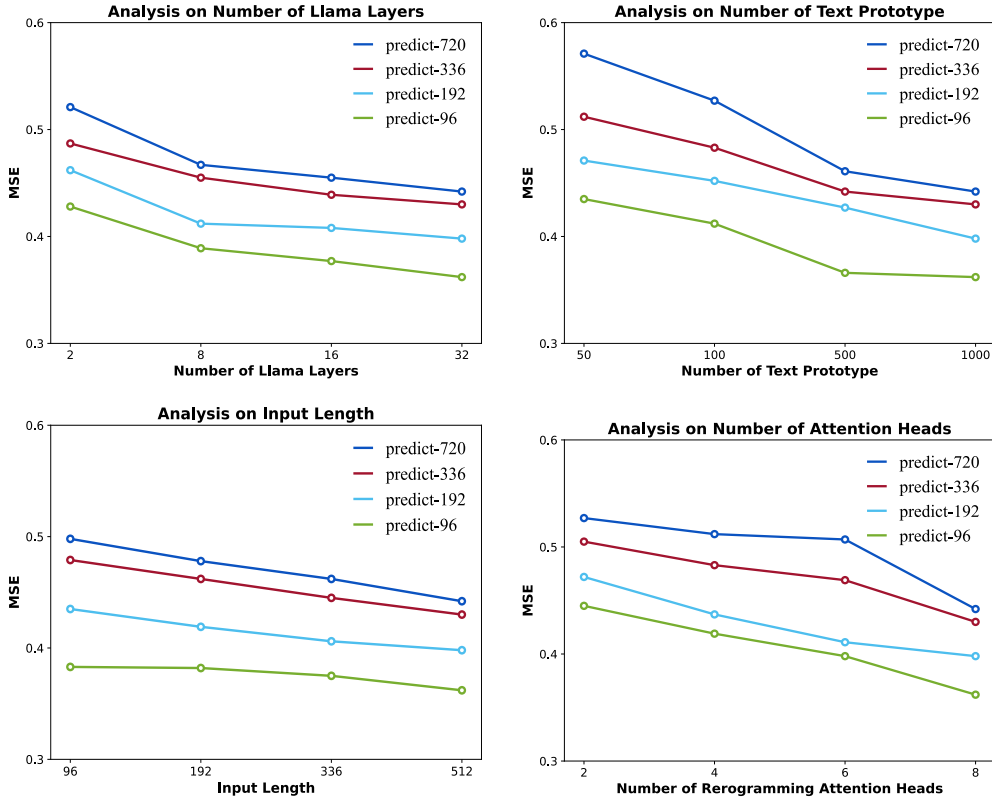


Figure 6: Analysis of hyperparameter sensitivity on ETTh1 dataset.

D LONG-TERM AND SHORT-TERM FORECASTING

D.1 LONG-TERM FORECASTING

By solely reprogramming the smallest Llama model while keeping it intact, TIME-LLM attains SOTA performance in **36** out of 40 instances across eight time series benchmarks. This underscores the considerable potential of LLMs as robust and reliable time series forecasters. Furthermore, we benchmark the proposed method against other well-established baselines in Tab. 11. This comparison includes three notable statistical methods (AutoARIMA, AutoTheta, and AutoETS) (Herzen et al., 2022) and two recent time series models, N-HiTS (Challu et al., 2023b) and N-BEATS (Oreshkin et al., 2020). Remarkably, TIME-LLM secures SOTA performance across all cases, surpassing the second-best results by significant margins of over **22%** and **16%** in terms of MSE and MAE.

